

CS201 Midterm 1 Part B

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Olcay Taner YILDIZ

Abstract—Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted,

- (i) **The input data structure is not empty**
- (ii) **Time complexity of the algorithm does not matter**
- (iii) **You can not use extra data structures including arrays**
- (iv) **You can use any function given in the data structure**
- (v) **You can not modify the data structure contents**

I. QUESTION

Write a function that returns the Zibonacci numbers between A and B (both inclusive) as a singly linked list. The first three Zibonacci numbers are $Z_0 = 0$, $Z_1 = 1$, $Z_2 = 1$ and the general formula is $Z_n = Z_{n-1} + Z_{n-3}$. Assume that $A \geq 2$ and $B > A$.

```
LinkedList zibonacci(int A, int B)
```

II. QUESTION

Write a function that checks whether the original singly list contains at least one increasing sequence of length $k > 1$ **exactly**.

```
boolean isIncreasingOfSizeK(int k)
```

III. QUESTION

Write the method

```
boolean checkString(int[] s)
```

which checks if the array s is of the form $1^m 2^m$ without counting any characters and using an external array based stack.

- You can assume the array consists of only numbers between 1 and 2 and there is at least one from each of them.
- You are only allowed to use pop, push, isEmpty functions.
- You should use one single external stack.
- You can not modify s .
- Beware, you should return false for the strings 21, 1212, or 112212.
- Implementation should be done on array based stack.

IV. QUESTION

Write a new stack constructor

```
Stack(Stack s, int p, int q)
```

which constructs a new stack by copying elements from stack s indexed between p and q (both inclusive).

- Assume that $q \geq p$.
- The elements of the constructed stack should be new, not the elements of s .
- The first element at the top of the stack s has index 0.
- You are not allowed to use any stack methods and any external stacks.
- You are allowed to use attributes, constructors, getters and setters.
- You can not modify s .
- Implementation should be done on linked list based stack.